

Puthin Kumar Reddy Thallapalli

Data Analyst

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SUMMARY

- Data Analyst with 4-year experience in healthcare, e-commerce, and predictive analytics, leveraging Python, SQL, and R for data extraction, cleaning, and analysis.
- Expertise in developing machine learning models for sales forecasting, customer segmentation, and recommendation systems to drive business insights and optimization.
- Skilled in data visualization using Power BI, Tableau, and QlikView to create interactive dashboards and reports, presenting actionable insights to key stakeholders.
- Proficient in HIPAA compliance monitoring, ensuring secure patient data handling and automating compliance reporting with Python to streamline audit processes.
- Experienced in building and managing ETL pipelines and working with cloud technologies such as AWS (EC2, S3, Redshift) and Google Cloud (Big Query).
- Strong background in statistical analysis and data mining, utilizing libraries like Pandas, NumPy, Scikit-learn, and Stats models to derive insights and optimize decision-making.
- Agile and Scrum experience, collaborating effectively with cross-functional teams (marketing, IT, and product) to ensure business objectives are met through data-driven solutions.
- Skilled in data wrangling, data cleansing, and model evaluation techniques (e.g., RMSE, MAPE, F1 score) to ensure high-quality, accurate outputs.

SKILLS

- **Programming Languages:** Python, SQL, R
- **Data Analysis Tools:** Excel (Advanced: Pivot Tables, VLOOKUP, Power Query, VBA), R Studio, Jupiter, VS Code
- **Data Visualization:** Tableau, Power BI, QlikView
- **Databases:** MySQL, MS SQL Server, PostgreSQL, MongoDB, Firebase.
- **Cloud Technologies:** AWS (EC2, S3, Redshift), Azure (Azure Data Factory, Databricks, Azure Synapse), Google Cloud (Big Query)
- **ETL & Data Pipelines:** Talend, Apache Airflow, AWS Glue, Informatica, Azure Data Factory
- **Big Data & Distributed Computing:** PySpark, Databricks
- **Machine Learning Tools:** TensorFlow, PyTorch, Keras
- **Data Analysis Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, XGBoost, Stats models
- **Version Control:** Git, GitHub
- **Data Analysis Skills:** Supervised & Unsupervised Learning, Predictive Learning, Neural Networks, Model Evaluation and Tuning, EDA
- **Methodologies:** Agile, Waterfall, Scrum
- **Analytical Skills:** Data Mining, Data Wrangling, Statistical Analysis, Data Cleansing, Text Mining
- **Healthcare Tools & Technologies:** HIPAA, EMR, HER
- **Enterprise Systems:** SAP, SAS

WORK EXPERIENCE

Molina Healthcare

August 2024 – Present

Data Analyst

- Developed scalable ETL pipelines using PySpark and Azure Databricks, enabling efficient processing of large healthcare datasets for compliance monitoring.
- Automated compliance reporting workflows using Python and Azure Data Factory, reducing manual reporting efforts by 40 hours per month and improving accuracy across departments.
- Designed advanced Power BI dashboards and leveraged Excel VBA for dynamic patient data analysis, enabling timely identification of compliance issues.
- Conducted in-depth SQL analysis of audit logs to detect data access violations, enhancing data security and regulatory compliance efforts.
- Optimized query performance and data retrieval using Azure Synapse Analytics, improving accessibility for cross-functional teams.

Molina Healthcare

January 2024 – March 2024

Data Analyst - Internship

- Assisted in implementing a HIPAA compliance monitoring system to track and report on patient data access, ensuring adherence to privacy and security regulations.
- Extracted and analyzed audit logs using SQL, identifying potential data access violations and ensuring regulatory compliance.
- Supported the automation of compliance reporting using Python, reducing manual audit processing time by 30% and improving efficiency in tracking HIPAA-related activities.

Infosys BPM

November 2020 – July 2022

Data Analyst

- Developed a hybrid recommendation system, increasing personalized product recommendations by 15%, enhancing customer retention.
- Extracted and cleaned large datasets of user behavior, transaction records, and product information using SQL and Python (Pandas, NumPy) to ensure data integrity and prepare for model training.

- Designed and implemented recommendation algorithms using collaborative filtering and content-based filtering techniques, applying machine learning models to predict customer preferences and suggest personalized products.
- Developed visualizations in Power BI to track the system's performance, monitor user interactions, and measure the effectiveness of product recommendations.
- Fine-tuned recommendation algorithms by evaluating performance using precision, recall, and F1 score to ensure high-quality recommendations. Improved model accuracy by testing different features and tuning hyperparameters.
- Worked closely with the marketing and product teams to ensure the recommendation system aligned with business goals and user experience objectives.
- Employed Agile practices to iteratively build and improve the recommendation system, incorporating feedback from stakeholders and making continuous enhancements based on user testing.

Trigent

January 2020 – October 2020

Data Analyst

- Designed and developed a predictive sales forecasting model to optimize inventory management and marketing strategies for an e-commerce platform.
- Utilized Python (Pandas, Scikit-learn) and SQL to preprocess, clean, and analyze large datasets, including historical sales data, customer demographics, and website traffic patterns.
- Built and deployed time-series forecasting models (ARIMA, Random Forest) to predict monthly sales, improving inventory accuracy by 20% and reducing stockouts.
- Conducted Exploratory Data Analysis (EDA) to identify seasonal trends, customer behavior patterns, and external factors influencing sales performance (e.g., promotions and holidays).
- Created interactive dashboards in Tableau to visualize KPIs, sales trends, and forecasting results, providing actionable insights for stakeholders and business teams.
- Collaborated with cross-functional teams (marketing, operations) using Agile methodology to iteratively refine forecasting models and integrate feedback into the final solution.
- Analyzed the performance of the predictive models and performed model evaluation (e.g., RMSE, MAPE) to fine-tune predictions and ensure their accuracy.

PROJECTS

Healthcare Claims Fraud Detection

- Designed and implemented an anomaly detection system using unsupervised machine learning models (Isolation Forest, DBSCAN) to identify fraudulent healthcare claims, improving fraud detection rates.
- Developed and optimized ETL workflows in Apache Airflow, automating data ingestion from multiple sources (claims data, patient records), ensuring real-time fraud monitoring.

Customer Churn Prediction for E-Commerce

- Built a predictive churn model using Logistic Regression and XGBoost, analysing customer behaviour patterns to identify high-risk customers and enhance retention strategies.
- Integrated Snowflake and Tableau to create an interactive churn analysis dashboard, enabling business teams to take data-driven actions for reducing churn.

EDUCATION

- **University of Alabama at Birmingham**
Master of Science in Computer Science - Birmingham, Alabama
- **Madanapalle Institute of Technology & Science**
Bachelor of Technology in Computer Science - Andhra Pradesh, India